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Paleomagnetism of Rocks From Burin Peninsula, Newfoundland: Hypothesis of Late Paleozoic Displacement of Acadia Criticized

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The posttectonic St. Lawrence Granite (360 ± 5 Ma) and associated subvertical dikes have two magnetizations, *N* and *H*. *N* points north with an intermediate upward inclination (declination, 005°; inclination, −50°; alpha (95), 10°; paleopole 12°N, 120°E). We consider *N* to have been acquired close to the time of initial cooling. The *H* magnetization points south with low inclination. *H* is also the dominant magnetization in the nearby Permo-Carboniferous Spanish Room Formation. *H* is considered to be an overprint acquired chemically during the Kiaman Reversed Superchron (Late Carboniferous and Permian). The *N* magnetization corresponds to a paleolatitude of 31 ± 7°S, very different from the near-equatorial values expected from the commonly accepted Late Devonian and earliest Carboniferous (Tournaisian) reference paleofields for cratonic North America. This is generally consistent with the hypothesis of a southerly displaced Acadia proposed by Kent and Opdyke, although the displacement is much greater (30° rather than 15°). However, we show that all inclinations observed from Middle to Upper Devonian and Tournaisian rocks from the craton are not significantly different either from the Kiaman paleofield, or from the *H* overprints observed from Newfoundland. Some of these cratonic studies are not very detailed, but it is noteworthy that the declination observed in the most comprehensive of them (Catskill red beds) also agrees with the Kiaman paleofield and with the *H* overprints from Newfoundland. Declinations in some other cratonic studies show small departures of only a few degrees (with one exception based on limited sampling and two from the Colorado Plateau which may have been rotated) which could be caused by experimental inaccuracies. Moreover the directions obtained from cratonic studies are 97% reversed, as they are in rocks with Kiaman magnetizations. We suggest, therefore, that the commonly accepted Late Devonian to earliest Carboniferous reference paleofield for North America is either entirely Kiaman in age or has been seriously affected by Kiaman overprinting. Consequently, the 30° of southerly displacement which the *N* magnetization of the St. Lawrence Granite apears to indicate, and the 15° invoked by Kent and Opdyke, we consider to be illusory. Three Late Precambrian rock units (623 to 606 Ma) have scattered declinations but similar inclinations yielding paleolatitudes of about 35°. The scattered declinations presumable were caused by local pre-Carboniferous relative rotations. There is little or no evidence of Kiaman overprinting in Precambrian rocks, indicating that overprinting is comparatively superficial and is probably related to Late Paleozoic groundwater circulation.

INTRODUCTION

The main purpose of this paper is to describe new paleomagnetic results from the Late Devonian–Early Carboniferous St. Lawrence Granite of the Avalon Zone of eastern Newfoundland, and to compare them with results from rocks of comparable age from cratonic North America in order to determine if there has been any motion between the Avalon Zone and the craton.

The Avalon Zone, of which the Burin Peninsula is part, extends intermittently from Newfoundland to Florida. Its rock sequences have been compared to similar sequences of North Africa and western Europe [*Schenk*, 1971; *Rast et al.*, 1976; *Strong*, 1978], and it could therefore be a fragment of the eastern continents left behind by the Mesozoic opening of the Atlantic [*Wilson*, 1966]. Avalon is of special importance for the evolution of the Appalachians because it is the oldest (late Precambrian) terrane east of the Proto-Atlantic (Iapetus) ocean, an earliest Paleozoic ocean extended eastward from the Precambrian platform of western Newfoundland across what is now central Newfoundland. Iapetus, the width of which is undetermined, separated the Avalon Zone from North America until it closed in the Ordovician.

Later motions between Avalon and the platform of western Newfoundland, may also have occurred. Two have been suggested. First, Newfoundland may have rotated anticlockwise, presumably during the Mesozoic or Cenozoic [*Wegener*, 1924, p. 60]. Paleomagnetically this would manifest itself as an anticlockwise rotation of the paleomagnetic declinations (paleonorth or paleosouth direction depending on whether the paleofield was normal or reversed) relative to those from the cratonic North America. Some paleomagneticians have discussed the possibility of rotation favorably [*Black*, 1964], others unfavorably [*Robertson et al.*, 1968; *Deutsch and Rao*, 1977]. Second, mid-Carboniferous motions have been invoked by *Kent and Opdyke* [1978] on the grounds that inclinations (from which paleolatitudes are determined) in Lower Carboniferous and Upper Devonian rocks from north central Appalachians, which include the Avalon Zone and which was referred to by Kent and Opdyke as "Acadia," are higher than those from the North American craton. They argued that Acadia was then situated about 1500 km farther south relative to the craton than at present (Figure 1). Paleolatitudes for the Late Carboniferous agree across the orogen. Hence Kent and Opdyke argued that the motions were related to the mid-Carboniferous Alleghanian orogeny. Paleomagnetic determinations of the ancient geomagnetic field (paleofield) from the Avalon Zone could assist in mapping these motions, provided, of course, that an adequate set of coeval determinations are available from cratonic North America.

The Kent-Opdyke proposal has been criticized by *Roy* [1982] and *Roy and Morris* [1983], on the grounds that comparisons between cratonic and Appalachian results are of

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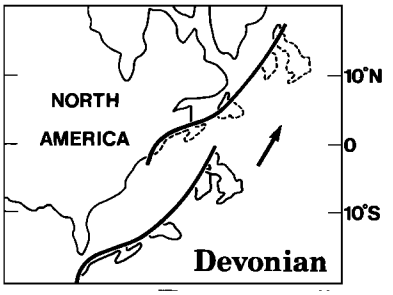


Fig. 1. Hypothesis of displaced Acadia [*Kent and Opdyke*, 1978]. Reproduced with permission.

doubtful significance, because the reference paleofield for cratonic North America is poorly known and may contain systematic unremoved overprints acquired during the Late Carboniferous and Permian (Kiaman). *Kent and Opdyke* [1978, p. 4447] considered the possibility of Kiaman overprinting, but abandoned it on the grounds that there was no experimental evidence for it, and because of the good agreement among widespread cratonic results obtained from different rock types. A further objection that could be raised against their hypothesis of Alleghanian megashear is the absence of paleomagnetically observed rotations in Lower Carboniferous and Upper Devonian rocks of the Appalachians. Large rotations are endemic in other zones of continental shear and collision [*Beck*, 1976; *Irving*, 1983], and the fact that they have not been seen in the Appalachians argues against the idea of Alleghanian megashear.

The cratonic remagnetization hypothesis of Roy and Morris has been greatly strengthened by a recent study of Viséan redbeds (the Deer Lake Group) of western Newfoundland [*Strong and Irving*, 1983; *Hiscott and Coniglio*, 1983, p. 125; *Irving and Strong*, 1984]. These rocks belong to the craton because they rest directly on Precambrian basement of the Canadian Shield of the west Newfoundland platform. The Deer Lake study shows two things. First, the rocks possess an older prefolding Viséan magnetization which yields a paleolatitude (20 ± 6°S), differing by 20° from those obtained from Lower Carboniferous cratonic rocks of the United States. Second, the Deer Lake redbeds possess a younger Kiaman (Late Carboniferous and Permian) overprint which yields a paleolatitude which does not differ significantly from those obtained for Lower Carboniferous rocks of the craton, indicating that the magnetism of the latter is, as Roy and Morris argued, predominantly Kiaman. These new data do not preclude the possibility of motion on the scale of hundreds of kilometers, which would not be detectable paleomagnetically, but they do show that the displacement of 1500 km invoked

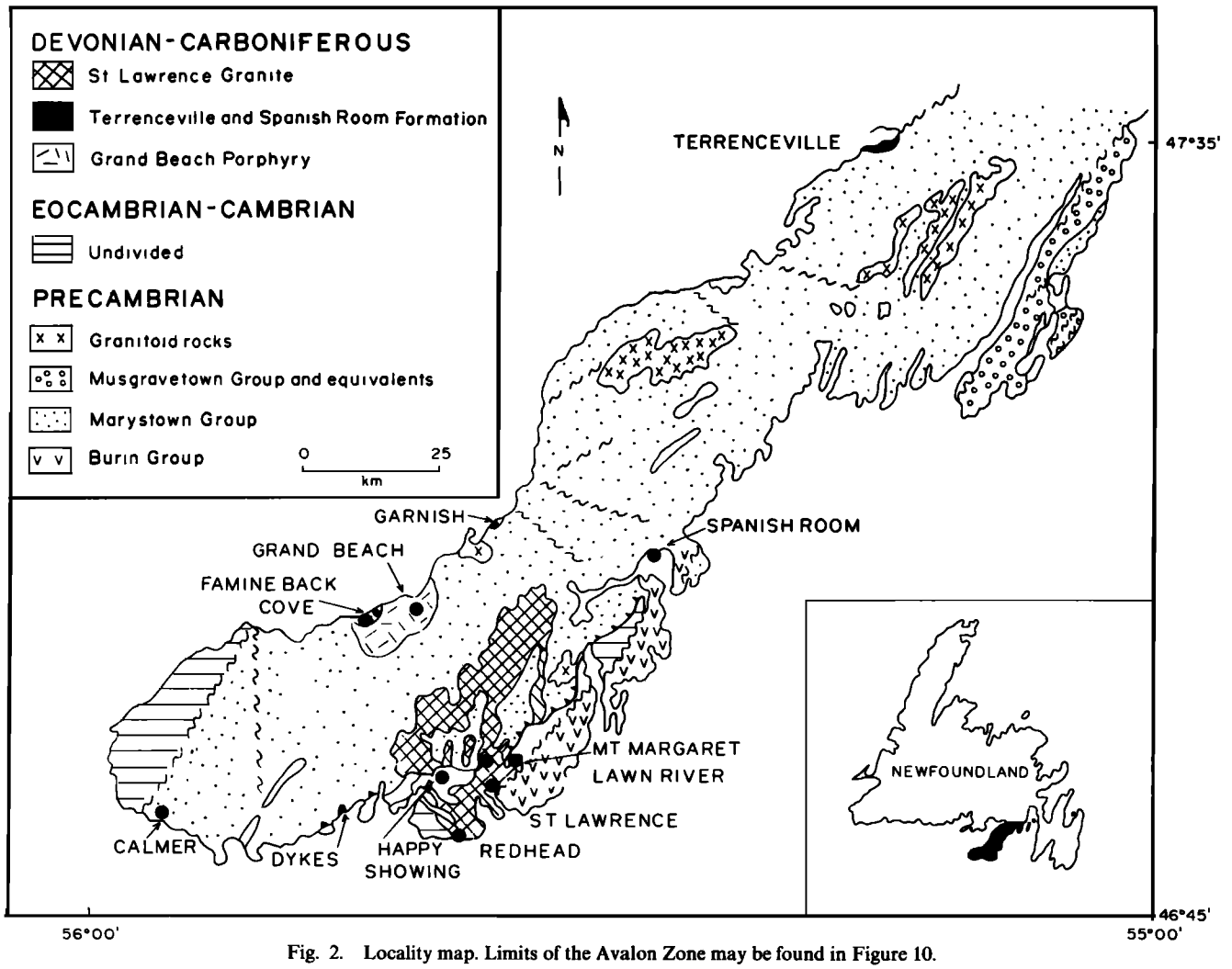


Fig. 2. Locality map. Limits of the Avalon Zone may be found in Figure 10.

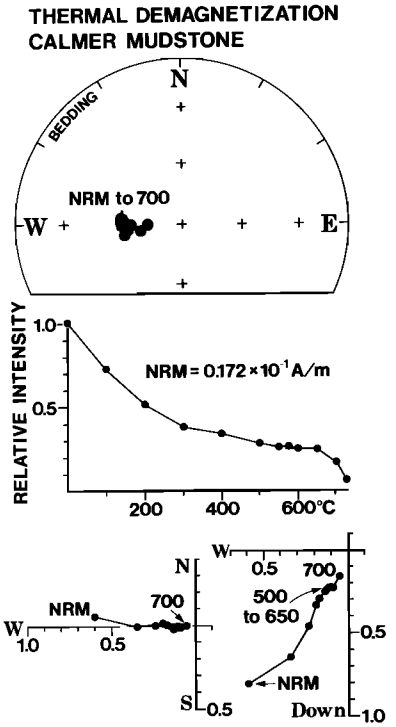


Fig. 3. Thermal demagnetization of red-purple Calmer mudstone. The direction and intensity changes above and orthogonal plots (normalized NRM = 1.0) below.

by *Kent and Opdyke* [1978] is unlikely to have occurred during or since the Viséan.

Since the original submission of this paper an abstract has appeared [*Kent and Opdyke*, 1984] describing a restudy of a Lower Carboniferous cratonic rock unit, the Mauch Chunk Formation of Pennsylvania. From their new study they now recognized that the predominant magnetization is indeed a Kiaman overprint, confirming the remagnetization hypothesis of *Roy and Morris* [1983]. There is also an underlying prefolding magnetization, which yields a paleolatitude of about 15°S, and which confirms our result from the Deer Lake Group. The possibility however remains that motions of the type invoked by *Kent and Opdyke* [1978] could have occurred in the earliest Carboniferous (Tournaisian) or latest Devonian, and it is with this question that we are now concerned.

Previous paleomagnetic work from the Avalon Zone has yielded a somewhat confused picture. *Buchan and Hodych* [1982] have argued that the paleomagnetism of the Ordovician Wabana and Bell Island Groups, if assumed to be original, indicates little or no displacement relative to North America since the Ordovician. If it is secondary and Devonian in age, then a displacement from the south would, they argue, be indicated. *Kent* [1982], in his study of the Terenceville Formation, observed a postfolding Kiaman overprint and a prefolding magnetization with intermediate inclination which he considered to be Late Devonian, the age he assumed for the beds. Palynological studies indicate that the beds are in fact Tournaisian [*Howie and Barss*, 1975]. Kent argued that the prefolding magnetization was consistent with the Alleghanian displacement invoked by *Kent and Opdyke* [1978]. Finally *Murthy* [1983*a*, *b*] has studied two sets of dikes, both of probable Late Devonian age. The first set, from near Deadmans Bay, has a stable southerly directed magnetization with shallow inclinations, which could be a Kiaman overprint (see ahead, Figure 17). The second set, from Bonavista Bay around Wesleyville, possesses a stable magnetization with reversals and moderate inclinations (Figure 16). Both sets also have a steeply inclined very soft magnetization which we consider to be of recent origin. Murthy reviewed the problem cautiously. He did not regard his results as strong evidence for or against displacement.

Our study focuses on some of the oldest and the youngest rocks of the Avalon Zone, especially the latter, in an attempt to detect post–Late Devonian movements. As we shall show, our attempt has not been unsuccessful.

Somewhat paradoxically, the magnetizations of most of the Precambrian rocks are comparatively simple, whereas those of the Late Paleozoic rocks are generally more complex. As we shall see, we believe this is caused by Kiaman overprinting which preferentially affected the more superficial rocks. Paleozoic rocks require detailed alternating field (af) or thermal demagnetization studies of each specimen (10 or more steps) to separate the magnetizations present. Separation of low and intermediate stability magnetizations have been obtained by calculation of the magnetization removed (vector differences) along linear trajectories in the orthogonal plots. Components of very high stability, when present, have been recognized by their stable end-points. We have preferred to get detailed results from comparatively small collections rather than less detailed results from larger collections.

PRECAMBRIAN RESULTS

The Marystown Group is the name given by *Strong et al.* [1978*b*] to the dominantly subaerial volcanic-sedimentary sequence underlying the southern Burin Peninsula (Figure 2). It has since been recognized throughout the peninsula [*O'Brien*,

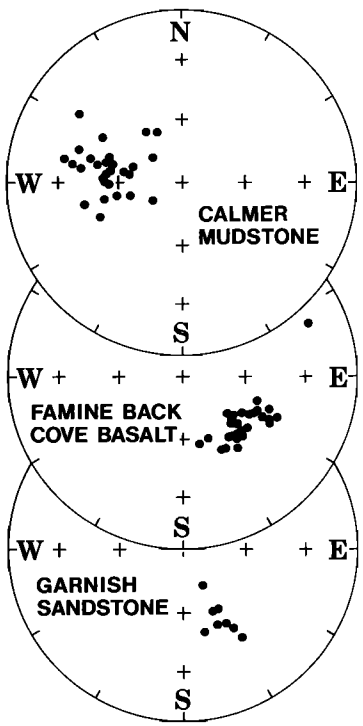


Fig. 4. Cleaned directions of magnetization observed in three units of the Marystown Group, all with respect to bedding. See footnote to Table 1 for bedding attitudes.

TABLE 1. Results From Marystown Group

Rock Unit	n	s	Declination, deg	Inclination, deg	k	Alpha (95)
1, Calmer mudstone	18*	28	282	54	38	6
2, Famine Back Cove basalt	15*	28	133	53	63	5
3, Garnish red sandstone	6*	11	151	53	85	7

Variable n is the number of cores, s is the number of specimens, declination and inclination are of the mean corrected for bedding tilt, k is Fisher's precision, and alpha is the half angle of the cone of confidence ($P = 0.05$). Bedding is as follows: (1) 30° at 270°, (2) flat, and (3) 25° at 180°. The compilation procedures are as follows: (1) average 550° to 650°C; (2) average 550° to 650°C, 21 specimens; average 20 to 80 mT, 7 specimens; (3) average 550° to 600°C.
*Unit weight is given to cores.

1979]. It is conformably or unconformably overlain by an Eocambrian-Cambrian sedimentary sequence. Within the sequence, several conformable units can be recognized. An ignimbrite unit has been dated by the U/Pb (zircon) method at 607(+8.0/−3.7) Ma, which is comparable to ages of other ignimbrites 623(+1.9/−1.7) and 606(+3.7/−2.9) Ma) from elsewhere in the Avalon Zone [*Krogh et al.*, 1983]. The group is dominated by silicic pyroclastic rocks, interbedded with basaltic flows and minor clastic sediments. The flows are subaerial and somewhat oxidized, with scoriaceous tops and filled vesicles. They have continental-tholeiite composition, with a local abundance of oxidized clinopyroxene and olivine phenocrysts. The sedimentary rocks were derived locally from volcanic rocks. Three units, and a fourth of probable Marystown age, were sampled.

Calmer mudstone. Eighteen cores were drilled from red-brown to purple mudstones at the village of Calmer (Figure 2). These mudstones occur as gently-dipping beds, up to a meter thick. There is a locally developed weak fracture cleavage which was avoided during drilling. The samples span 10 m stratigraphically. The rocks contain a single dominant magnetization, which, after correction for tilt, is directed to the west with moderate downward inclination. The orthogonal plots converge linearly to the origin (Figure 3). An unusual feature is that after heating at 700°C, over 10% of the magnetization remains unchanged in direction. Similar high temperature memory effects have been described in diorites by *Morel* [1981]. The directions after thermal cleaning are shown in Figure 4, and their overall mean is listed in Table 1. Basalts also occur at Calmer, but they were not sampled because precise bedding attitudes could not be obtained.

Famine Back Cove Basalt. Fifteen cores were drilled from flat-lying reddish amygdaloidal basalts. They are considered to be correlative with the Calmer basalts [*Strong et al.*, 1978*b*]. They occur on the beach east and west of Famine Back Cove, with an exposed thickness of 15 m. They were sampled through their thickness over a distance of 1 km. The magnetization of three cores is univectorial (Figures 5 and 6). It remains little changed in alternating fields up to 100 mT and temperatures up to 650°C, and it points to the southeast with intermediate downward inclination. The basalts have undergone intense deuteric alteration and hematite is the main carrier of the magnetization. In nine cores there is also a substantial (*PEF*) component, aligned close to the present earth's field, that is removed at low temperatures or after treatment in low alternating fields. Excellent end-points are found by both af and thermal demagnetization in these 12 cores. The remaining three cores have not only *PEF* and *SE* magnetizations, but also very small magnetizations sometimes pointing south sometimes north. These very small components could not be accurately isolated (Figure 7). The northerly ones could be overprints acquired at the time of intrusion of the St. Lawrence Granite (see below). The southerly ones could be Kiaman overprints, common in Newfoundland [*Kent*, 1982; *Irving and Strong*, 1984]. The directions of dominant *SE* magnetizations are plotted in Figure 4, and their mean is listed in Table 1.

Garnish red sandstone. The third unit sampled consists of

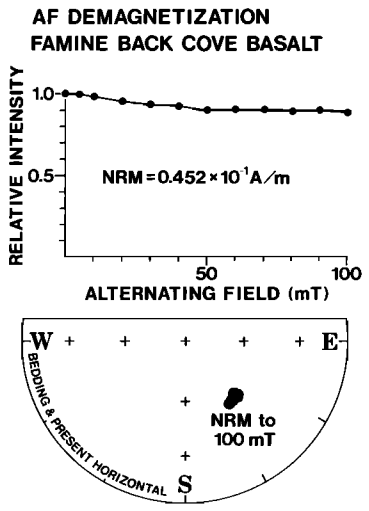


Fig. 5. Alternating field demagnetization of the Famine Back Cove red basalt.

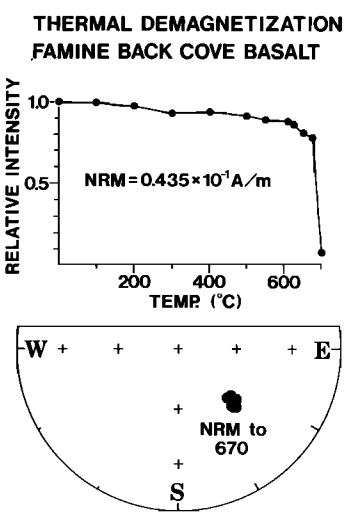


Fig. 6. Thermal demagnetization of Famine Back Cove red basalt, one magnetization.

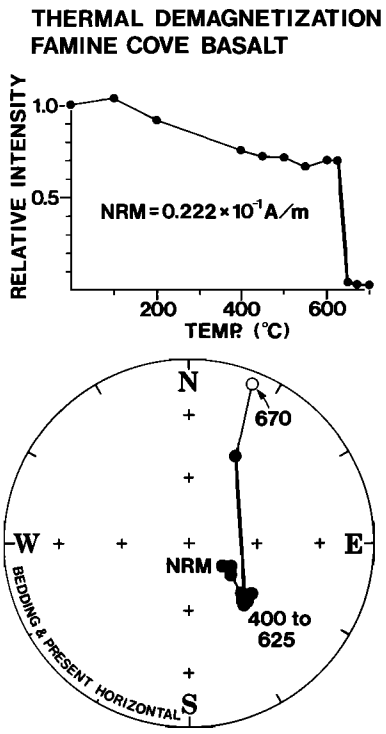


Fig. 7. Thermal demagnetization of Famine Back Cove red basalt showing initial migration away from present earth's field, and the presence, at high temperatures, of a small poorly defined northerly magnetization.

basalts and red sandstones exposed at Deepwater Point about 1 km south of Garnish (Figure 1). They are believed to comformably underlie the beds sampled at Calmer and Famine Back Cove. Like them they are gently dipping, with weak fracture cleavage, always avoided during sampling. Detailed demagnetization of 11 cores from the basalts failed to yield coherent magnetizations. The sedimentary rocks are red cross-bedded standstones and siltstones with intercalated horizons of conglomerates. The red sandstone horizon sampled overlies the basalt and fills fractures in the flows. After removal of *PEF* components, high blocking temperature *SE* magnetizations remained (Figure 8). As in the Calmer mudstone, appreciable magnetization is retained in some specimens at 700°C. The cleaned directions are plotted in Figure 4 and their mean listed in Table 1.

These three units of the Marystown Group have similar mean inclinations after correction for tilt of the beds (54°, 53°, 53°, Table 1, Figure 4) corresponding to paleolatitudes of about 35°. Before correction, the inclinations are 81°, 53°, and 72°, respectively. The better agreement after rather than before correction, indicates that the magnetization predates the tilting. The age of the tilting is not known precisely although it is probably a consequence of the Acadian orogeny when thrust-faulting and folding occurred. It is certainly pre-Carboniferous as shown by unconformable relations with Carboniferous sediments (the Spanish Room and Terenceville Formations, for example) and post-tilting intrusive relationships with the St. Lawrence Granite. Hence the magnetization is pre-Carboniferous. The low grade of the metamorphism and the high stability of the magnetizations would suggest that the magnetizations are late Precambrian, but this is by no means certain. The declinations of the Garnish red sandstone (151°) and the Famine Back Cove basalts (133°) are roughly similar, but that of the Calmer mudstone (282°) is different, the magnetization pointing west rather than southeast. Such differences in declination are common indicators of regional shear in orogenic belts [*Beck*, 1976], and presumably reflect rotation around vertical axes, which occurred possibly in the Acadian orogeny.

Precambrian paleopoles in the age range 600–620 Ma are not available from the Canadian Shield [*Roy*, 1983] so no comparisons can be made with results from the Burin Peninsula. Late Precambrian rocks of the Avalon Zone resemble those of the Moroccan Meseta, and it has been suggested that they were formerly in close proximity [*Schenk*, 1972; *Strong*, 1978]. Paleomagnetic evidence, summarized by *Morel* [1981], indicates that northwest Africa at about 600 Ma was close to the paleopole. The paleolatitudes we have observed (35°) are not consistent with this. Because of the uncertainties in intercontinental correlations in the Precambrian, this is an insecure conclusion.

Other Precambrian samples. The fourth sample site from Precambrian rocks was in pillowed basalt flows at Mount Margaret (Figure 2). These were originally assigned to the Burin Group [*Strong et al.*, 1978*b*], which uncomformably underlies the Marystown and is dated at 762 Ma [*Krogh et al.*, 1983]. On the basis of more detailed mapping, *Wilton* [1976] suggested that they are probably not part of the Burin and possibly belong to the Marystown Group. Three samples were taken from within the aureole of the St. Lawrence Granite, and all are hornfelsed. They have simple magnetizations directed downward (358, 69) quite different from those in the pluton and essentially parallel to the *PEF* (335, 71). Unblocking temperatures are distributed between room temperature and 550°C. The magnetization is probably of recent origin. Sedimentary rocks from Eocambrian sedimentary rocks

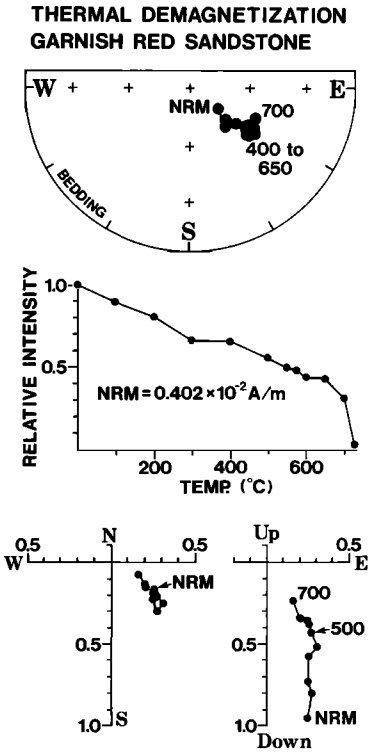


Fig. 8. Thermal demagnetization of Garnish red sandstone showing migration away from present earth's field and retention of appreciable magnetization after heating to 700°C.

within the granite contact aureole sampled at the Happy Showing Quarry (Figure 2) gave no coherent magnetization.

St. Lawrence Granite

Geology. The main body of this pluton consists of alkaline to peralkaline alaskite [*Teng and Strong*, 1976; *Strong et al.*, 1978*b*]. It was dated originally by a four-point Rb/Sr whole-rock isochron at 315 ± 10 Ma [*Bell and Blenkinsop*, 1975]. Subsequent more detailed studies have provided a 10-point isochron of 360 ± 5 Ma [*Strong et al.*, 1984], placing it at the Devonian-Carboniferous boundary [*Harland et al.*, 1982]. The pluton intrudes most stratigraphic units in the center of the southern Burin Peninsula (Figure 2) and cuts all deformation features which they exhibit (thrust faults, folds, cleavage); that is, the St. Lawrence Granite is clearly posttectonic. Its contacts are sharp, with chilled quartz-feldspar porphyry margins and hornfelsing of the country rock. Miarolitic cavities occur near contacts, and tuffisites (gas breccias) are common, indicating the shallow nature of the intrusion.

The rock consists of quartz, orthoclase and albite, with minor amounts of riebeckite, aegirine, biotite, fluorite, magnetite, and hematite. Alkali feldspar makes up to 30 to 60% of the rock, and is generally turbid with finely disseminated hematite. Although some joint surfaces are slickensided, their consistent mutually perpendicular orientations, together with the continuity into the country rock of undeformed cogenetic flourspar veins, indicate that the pluton has not been significantly deformed or tilted since solidification. Tilting is always an important problem in paleomagnetic studies of intrusions and we shall return to it later.

The pluton shows evidence of alteration at both high and low temperatures. At Happy Showing (Figure 2), there is progressive alteration towards the chilled contact with Eocambrian shales and siltstones, by which the quartz and K feldspar are replaced by calcite and albite. This "carbothermal" alteration was interpreted by *Strong* [1982] as a high-temperature effect, related to cooling and devolatilization of the granite, an interpretation subsequently supported by fluid inclusion and isotropic studies [*Strong et al.*, 1984]. Samples from the alteration zone plot on the same 360 Ma Rb/Sr isochron as the unaltered granite samples, showing that the high temperature alteration is coeval with the granite [*Strong et al.*, 1984]. At Red Head (Figure 2), oxidation occurs along joints and fractures. It is marked by diffusion bands of variable hematite concentration, which reflect repeated waves of oxidizing fluids diffusing into the granite from fractures. This alteration probably occurred at low temperatures because it is not accompanied by any significant hydrothermal alteration. It presumably resulted from deeply circulating groundwaters and could have been responsible for the Kiaman overprint described below. At the other two localities sampled (Figure 2), along the Lawn River and along the highway north of St. Lawrence, the alaskite has a medium-grained equigranular texture, typical of the center of the pluton. It varies from white to pink, depending on degree of alteration.

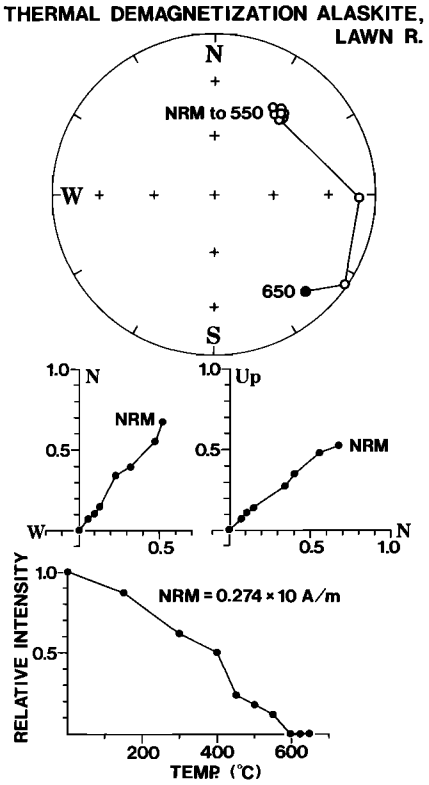


Fig. 9. St. Lawrence Pluton, *N* magnetization is alaskite of Lawn River. Thermal decay curve, directional changes, and orthogonal diagrams.

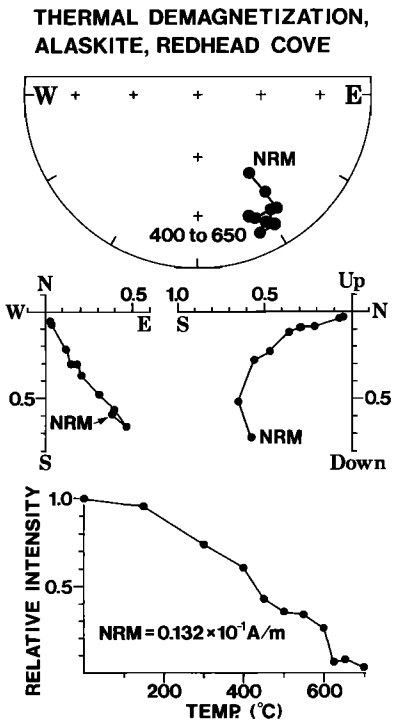


Fig. 10. St. Lawrence Pluton, *H* magnetization in alaskite at Redhead Cove.

Twenty hand samples were taken at the four places just noted (Figure 2). At Happy Showing the samples were confined to the quarry itself. At the other localities they are spaced over distances up to a kilometer. The samples were cored (1 core per sample) in the laboratory. One or two specimens were cut from each core.

Magnetizations of the alaskite. There are two stable magnetizations, one (*N*) directed toward the north with intermediate negative inclination, and another (*H*) directed almost horizontally to the south. A less stable *PEF* magnetization is also commonly present. In some specimens all three magnetizations occur, in others only two.

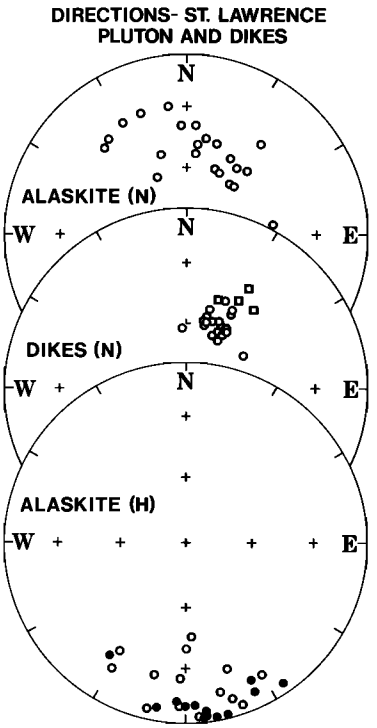


Fig. 11. *N* (above) and *H* (below) directions in St. Lawrence Pluton and *N* (middle) direction in diabase (open circles) and alaskite (open squares) of composite dikes. All with respect to present horizontal.

Thermal demagnetization results of a specimen with a dominant *N* and subordinate *H* magnetizations are shown in Figure 9. The unblocking temperatures are distributed predominantly below 600°C and the directions are constant up to that temperature. At 600°C the directions begin to migrate (the migration is progressive and not caused by random noise) to the south, showing that a small southerly directed magnetization is also present. Its intensity is very low. The essentially linear decay to the origin in the orthogonal plots shows that it does not mask *N*.

A typical *H* magnetization is shown in Figure 10. First there is a migration southward as a *PEF* component is removed, and then, between 400 and 650°, a good stable end-point with shallow inclination. The orthogonal plots have two-component signatures, terminated by linear decay to the origin. *N* is not present.

The directions of the *N* and *H* magnetizations observed in alaskite specimens from the pluton, are plotted in the upper and lower parts of Figure 11. The corresponding statistics, by locality, and by specimens, are given in Table 2 and paleopoles in Table 3. The directions are somewhat scattered but the groupings are clearly significant. The means of *N* and *H* are 120° of arc apart, not antiparallel. *H* is nearly horizontal with an upward inclination of −7°, compared with −51° for *N*. A considerable time lapse between the acquisition of *N* and *H* is required to explain their angular difference. Therefore, although *N* and *H* probably record a reversal of the field, they probably do not record one that occurred during a short interval of time, that is, during the cooling of the pluton. The demagnetization characteristics themselves provide no firm evidence of which is the older. However, although the *N* magnetization sometimes has unblocking temperatures above 600°C, it generally has "magnetite" unblocking temperatures (500°–600°C), whereas the *H* magnetization, when present, persists up to 650 or 675°C, the unblocking temperature of hematite (compare Figures 9 and 10). Since magnetite is an original magmatic mineral, it seems reasonable to associate the *N* magnetization with the original cooling of the pluton,

TABLE 2. Results From St. Lawrence Granite, Composite Dikes and Spanish Room Formation

Locality	*n*	*s*	Declination, deg	Inclination, deg	*k*	Alpha
1, Happy Showing, *N*	4	5*	339	−34	33	14
2, Happy Showing, *H*	3	6*	178	−26	17	17
3, Lawn River, *N*	5	9*	032	−56	16	14
4, Lawn River, *H*	4	4*	196	−12	05	48
5, Redhead, *N*	5	7*	016	−51	55	08
6, Redhead, *H*	9	11*	172	+02	27	09
7, St. Lawrence village, *N*	1	1	320	−33	...	...
8, St. Lawrence village, *H*	7	7*	174	−04	11	19
9, Alaskite mean by specimens, *N*	15	22*	008	−51	11	10
10, Alaskite mean by specimens, *H*	23	28*	177	−07	11	09
11, Composite dikes, diabase mean, *N*	11*	20	025	−58	92	05
12, Dikes, remagnetization circles, *S*	11	20	210	60	...	...
13, Composite dike alaskite, *N*	4*	4	031	−42	107	09
14, Composite dike, alaskite, *H*	1	1	210	08	...	...
15, Grand average alaskite and dikes, *N*	18		005	−50	14	10
16, Spanish Room Formation (horizontal)	19	30*	171	−01	27	05
17, Spanish Room Formation (bedding)	30*		171	−04	27	05

See Table 1 for headings. Except for entry 17, all directions are given relative to present horizontal. *N* and *H* are the magnetizations described in the text. In entry 15 unit weight is given to the 15 cores (15 hand samples, 22 specimens) of entry 9, the two diabase dikes and alaskite of the composite dikes each unit weight, and is considered to be the best summary of *N*. The experimental procedures follow, using these abbreviations: RM, magnetization removed in stated demagnetization interval; ave., average followed by demagnetization range in which the average is made. Entry 1, RM 600°–650°C (1), ave. 550°–650°C (2), 50 mT (2); entry 2, ave. 400°–600°C; entry 3, ave. 500°–650° (5), RM 10–60 mT (2), ave. 20–40 mT (2); entry 4, RM 100°–550°C; entry 5, RM 400°–550°C (5), ave. 500°–600°C (2); entry 6, RM 550°–600°C (1), ave. 400°–600°C (10); entry 7, 650°C; entry 8, RM 300°–550°C (2), ave. 300°–600°C (5); entry 11, RM 10–80 mT; entry 12, average of intersections of 20 remagnetization circles; entry 14, 650°C; entry 15, 650°C; entry 15, ave. 550°–650°C (26), RM 550°–650°C (4).

*Unit weight.

TABLE 3. Paleopoles

Unit and Location	Latitude	Longitude	*dm*, deg	*dp*, deg
1, Calmer mudstone (46.9°N, 57.9°W)	32°N	130°W	8	6
2, Famine Back Cove basalt (47.1°N, 55.6°W)	24°N	91°W	7	5
3, Garnish red sandstone (47.1°N, 55.5°W)	53°N	31°W	10	7
9, St. Lawrence pluton, *N* (46.9°N, 55.5°W)	11°N	118°E	13	9
15, St. Lawrence pluton and dikes, *N* (46.9°N, 55.5°W)	12°N	120°E	13	9
10, St. Lawrence pluton, *H* (46.9°N, 55.5°W)	47°N	129°E	9	5
16, Spanish Room Formation, hor. (47.2°N, 55.1°W)	43°N	137°E	5	3
17, Spanish Room Formation, bedding	42°N	136°E	5	3

The left-hand numbers correspond to entries in Tables 1 and 2.

and the *H* magnetization with its later low-temperature alteration.

Magnetization of composite dikes. Dikes outcrop in several localities on the southern coast of the Burin Peninsula between Red Head and Roundabout (Figure 2). They posttectonically intrude sheared and faulted rocks of the Marystown Group close to the western margins of the pluton, and range in width from 50 cm to about 10 m. All dikes are within 5° of vertical. Typically they have centers of alaskite and margins of diabase, each containing inclusions of the other, indicating the coexistence of mafic and silicic melts. The mafic margins contain unaltered plagioclase and pyroxene phenocrysts. The felsic centers consist primarily of quartz, K feldspar, and minor plagioclase, and are chemically identical to the alaskite of the St. Lawrence Granite. Miarolitic cavities occur. The dikes postdate all structures in the rocks which they intrude, and this, and their chemical affinities, indicate that they are offshoots of the St. Lawrence Granite.

Cores were drilled from two dikes. Eight cores from the diabase at the margin of one of them have very soft *PEF* magnetizations with weak indications of very small upward pointing more stable magnetizations which could not be accurately separated. Eleven diabase cores (20 specimens) from the other margin of the same dike and from the second dike all have three identifiable magnetizations. A typical result is shown in Figure 12. The initial direction is down and to the north. During af demagnetization it changes swiftly to a *N* direction as a large soft *PEF* magnetization is removed. There is then an apparent end-point, between about 10 and 30 mT, but at higher fields migration continues to the northeast along a great circle. The migration is not complete. The orthogonal diagrams show a linear segment corresponding to the removal of *PEF* magnetization, followed by a linear segment recording the removal of *N*. The latter segment decays almost but not quite to the origin, showing that a component with very high remanent coercive force remains untouched in 100 mT. Migration along great circles in high fields is found in all 20 specimens but never following the same path. The paths intersect at a point essentially antiparallel to the mean *N* magnetization of the dikes (compare entries 11 and 12 of Table 2).

Evidently the basaltic portions of the composite dikes contain three components. A large *PEF* magnetization, a large well-defined *N* magnetization, and a weak well-defined magnetization (called *S*) directed antiparallel to *N*. Attempts to isolate *S* by thermal demagnetization failed because at 300°C specimens fractured by transformation of zeolites. The directions of *N* are plotted in the central part of Figure 11 and their average given in Table 2. The mean does not differ significantly from that of the alaskite. No *H* magnetization was observed.

Alaskite cores from the composite dikes yielded *N* and *H* magnetizations (entries 13 and 14 of Table 2). Inclinations of *N* are somewhat shallower than in the diabase (Figure 11).

Samples were also taken from the Grand Beach Complex (Figure 2) but no coherent magnetizations were isolated.

SPANISH ROOM FORMATION

The Spanish Room Formation [*Strong et al.*, 1978*b*; *Laracy and Hiscott*, 1982] consists mainly of gently dipping porous red-brown sandstones, with abundant plant fragments and green reduction spots. Rare calcareous mudstones, and limestone caliche and conglomerate are interbedded with the sandstone. Its age is not accurately known. Plant fossils bear a

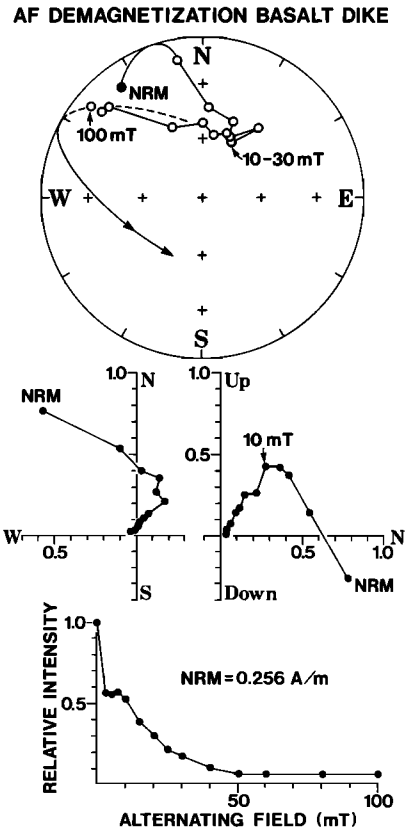


Fig. 12. Three components in composite dike. Direction (above) and intensity (below) changes, and orthogonal plots (center).

superficial similarity to those of the Terrenceville Formation 30 km to the north (Figure 2). Microfossils in shales of the underlying steeply dipping Cashel Cove Formation [*Laracy and Hiscott*, 1982] have been tentatively interpreted as Tournaisian in age, similar to those of the Terrenceville Formation [*Strong et al.*, 1978*b*]. *Laracy and Hiscott* [1982] suggested, by comparison with other Carboniferous strata of Atlantic Canada, that the Spanish Room Formation is Viséan to Lower Namurian. However its upper limit cannot be defined stratigraphically and it could be as young as Permian.

Nineteen cores were taken spanning 15 m stratigraphically. All have a predominant *H* magnetization. The thermal decay curve typically has two steps, the direction being constant throughout. Although there is some scatter, orthogonal plots show linear decay to the origin (Figure 13). In four specimens however, the directions migrated at 650° and 670°C by variable amounts to the north, and the decay is essentially linear but not always to the origin (Figure 14). Evidently a feeble northerly directed magnetization is occasionally present but it cannot be isolated. It could be an *N* magnetization with intermediate inclination, in which case the beds are Carboniferous rather than Permian in age. All observed directions are shown in Figure 15 and summarized in Tables 2 and 3.

ORIGIN OF *H* MAGNETIZATION

The *H* magnetizations are near horizontal and hence were formed near the paleoequator. *H* is widespread in the St. Lawrence Granite and makes up essentially the entire magnetization of the Spanish Room Formation. It is rare or absent from Precambrian rocks and from the composite dikes. *H*-type magnetization has been observed elsewhere in Late Devonian and Early Carboniferous rocks of Newfoundland, in the Terenceville Formation [*Kent*, 1982], probably in the Deadman's Bay dikes [*Murthy*, 1983*b*], and in the Deer Lake Group [*Irving and Strong*, 1984]. In the alaskite of the St. Lawrence Granite, *H* was probably produced by low temperature alteration, an alteration which involved the deposition of hematitic diffusion bands. *Gall* [1983] has shown, from textural studies in the Deer Lake Group, that the red hematite cement was deposited as a result of groundwater circulation long after original deposition and after the formation of oxidized grain coatings. The Spanish Room Formation is conspicuously porous and is a good candidate for similar postdepositional chemical processes.

The *H* magnetizations from Newfoundland are, as we shall show below (Figure 17), parallel to the Kiaman (Late Carboniferous and Permian) field, and are likely to be of this age. They appear to be further examples of the latest Paleozoic secondary magnetizations which are found in several continents [*Creer*, 1968]. Creer argued that these were acquired by the dehydration of iron hydroxides and oxyhydroxides during deep tropical weathering. This deep and prolonged weathering

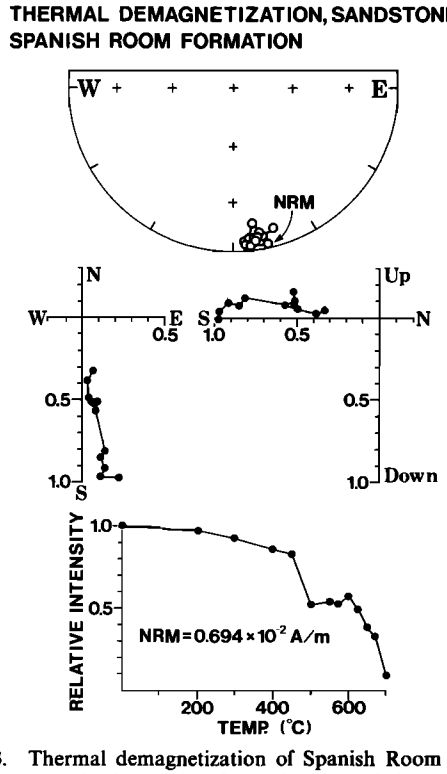


Fig. 13. Thermal demagnetization of Spanish Room Formation, single *H* magnetization. Direction (above) and intensity (below) changes, and orthogonal plots (center). The low intensity after heating is plotted on the intensity diagram but not on the others because the direction is wild. With respect to present horizontal.

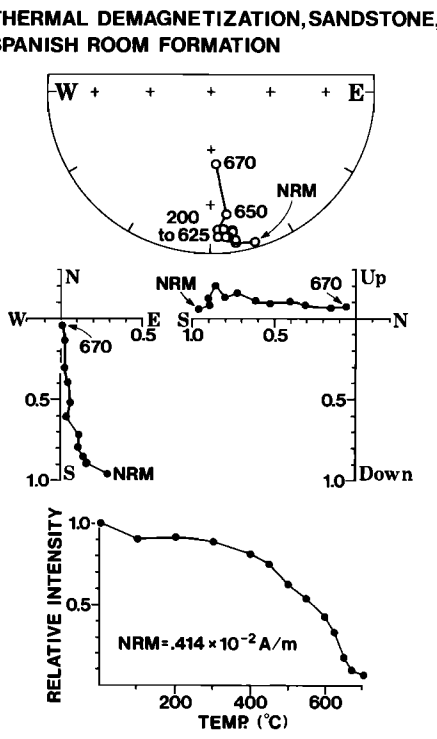


Fig. 14. Thermal demagnetization at Spanish Room Formation, dominant *H* component with possible indeterminate northerly magnetization. Direction (above) and intensity (below) changes, and orthogonal plots (center). With respect to present horizontal.

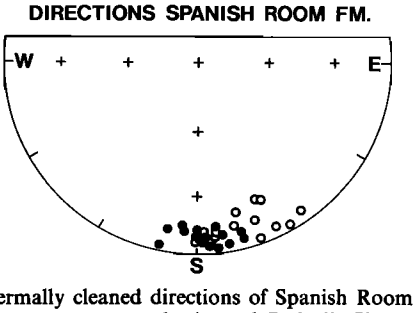


Fig. 15. Thermally cleaned directions of Spanish Room Formation with respect to present horizontal. Beds dip 7° at 217°.

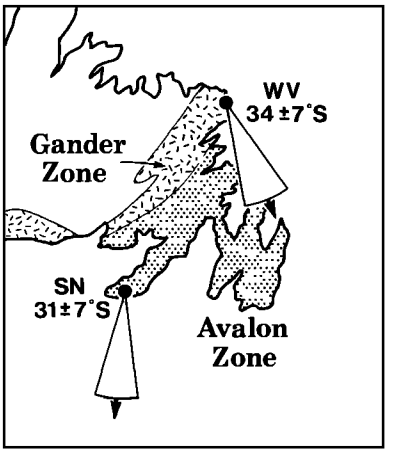


Fig. 16. Paleolatitudes and rotations determined from Late Devonian rocks of southeastern Newfoundland. The arrows point to paleosouth. SN, the *N* magnetization of the St. Lawrence Pluton; WV, the southeasterly magnetization of the Wesleyville dikes [*Murthy*, 1983*a*]. Details in Table 4.

presumably occurred because of the unparalleled continentality at the close of the Paleozoic. *Hedley* [1968] has shown, in controlled laboratory experiments, that such processes can produce stable remanent magnetizations. Since Newfoundland has not been deeply buried since the Carboniferous much of the present surface would have been accessible to such processes at that time. Because of their low porosity, Precambrian rocks and the diabase of the composite dikes have largely escaped the effect. Precambrian rocks would then have been more deeply buried below the zone of tropical weathering.

TECTONIC SIGNIFICANCE OF LATE PALEOZOIC RESULTS

The Kent-Opdyke hypothesis predicts that the Burin Peninsula, as part of Acadia, was about 1500 km south of its present position relative to cratonic North America in the Late Devonian and Early Carboniferous (Figure 1). We begin evaluating their proposal by first considering what the paleomagnetic results tell us about the paleolatitude of the southeastern Newfoundland at the end of the Devonian. We then make comparisons with results, first from elsewhere in Acadia, and then from cratonic North America. In order to ensure that results are applicable to their hypothesis the discussion will be confined to results from rocks of Middle and Upper Devonian, and earliest Carboniferous (Tournaisian) age.

Murthy [1983*b*] has studied subvertical diabase dikes (with 5° of vertical) from Bonvista Bay near Wesleyville in the Gander Zone (Figure 16). He considered them to have been intruded between 400 and 330 Ma, based on bracketing intrusive relationships of radiometrically dated granitoid plutons. Two of the dikes were dated by the $^{40}Ar/^{39}Ar$ method at 371 ± 8 and 373 ± 9 Ma, comparable to the age of the St. Lawrence Granite; we consider these to be good estimates of the age of the magnetization. The paleolatitudes of the two are in excellent agreement, $34° \pm 7°$S and $31° \pm 7°$S respectively (Figure 16). Paleosouth (the paleofield was reversed so the directions point to the ancient south pole) is $158° \pm 12°$ east of present north, compared with $185° \pm 12°$ for the St. Lawrence Granite. The difference indicates that there could have been some post-Devonian clockwise rotation of the Burin Peninsula relative to the northeast Gander Zone, occurring perhaps during sinistral shear along the Gander Zone [*Hanmer*, 1982], or as a consequence of slip along the Dover-Hermitage Bay fault which, according to *Blenkinsop et al.* [1976], may have been active as recently as 300 Ma. For the present discussion, the important result is that southeastern Newfoundland appears to have had a latitude of 30°–35°S at the end of the Devonian.

The agreement between results from subvertical dikes of Wesleyville (Bonavista Bay) and the posttectonic St. Lawrence Granite (with its associated subvertical dikes) confirms what is evident from field relationships (see above) that the St. Lawrence Granite has not been appreciably tilted since it was emplaced. In addition, the fact that the *H* magnetization (SH) matches so well the Kiaman field (Figure 17) indicates no tilting since the Permian. As noted earlier, postemplacement tilting of intrusive bodies is always a very real possibility, but in the case of the St. Lawrence Granite there is good evidence for many sources that differential internal tilting has been small, and that general tilting almost certainly has not exceeded 5°.

We now consider how well the Late Devonian–earliest Car-

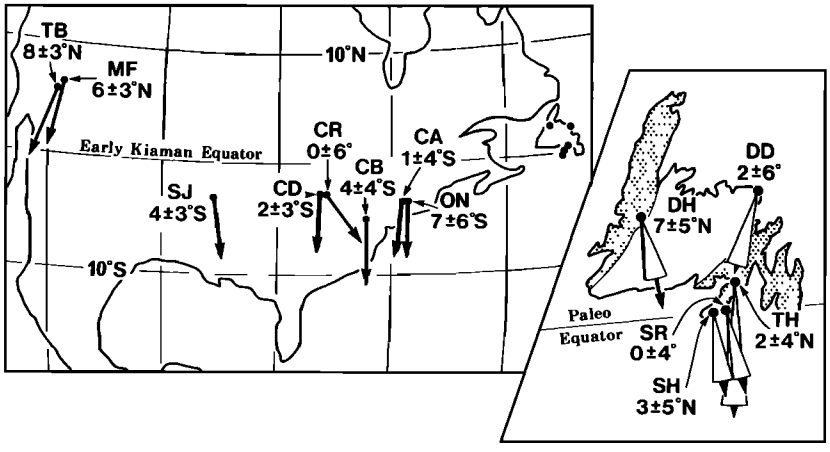


Fig. 17. Paleolatitudes and rotations observed in Middle and Upper (mainly upper) Devonian rocks of cratonic North America compared with the early Kiaman grid. In the inset, overprints observed in Carboniferous and Upper Devonian rocks of Newfoundland are shown. Note how well they accord with the position of the early Kiaman paleoequator. The western Newfoundland platform and the Avalon Zone are shaded. Details in Table 4.

TABLE 4. Comparisons With Early Kiaman Grid

Symbol	Rock Unit and Location	Rock Age	R, %	Mean Declination, deg	Inclination, deg	Alpha, deg	PLAT	RPD, deg	RR
SN	St. Lawrence Granite, *N* magnetization (46.9°N, 55.5°W)	Du	0	185 ± 12	50*	10	31° ± 7°S	−31 ± 8	−6 ± 13
WV	Wesleyville dikes, *SE* magnetization (49.1°N, 53.7°W)	Du?	20	158 ± 12	54	9	34° ± 7°S	−32 ± 8	22 ± 13
DD	Deadmans Bay dikes, *H* magnetization (49.2°N, 53.8°W)	C?	100	203 ± 8	4	10	2° ± 5°S	5 ± 6	−19 ± 9
SH	St. Lawrence Granite Overprint (46.9°N, 55.5°W)	Du	100	177 ± 7	−7	11	3° ± 4°N	3 ± 5	2 ± 8
SR	Spanish Room Formation, hor. (47.2°N, 55.1°W)	C-P	100	177 ± 4	−1	5	0° ± 2°	0 ± 4	8 ± 5
DH	Deer Lake Group overprint, hor. (49.0°N, 57.2°W)	C1	100	168 ± 8	−13	10	7° ± 4°N	−5 ± 5	10 ± 9
TH	Terenceville Formation, overprint (47.7°N, 54.7°W)	C1	100	186 ± 6	−3	7	2° ± 3°N	2 ± 4	−6 ± 7
SJ	St. Joe Limestone (Tournaisian)	C1	100	147 ± 5	9	6	4° ± 3°S	1 ± 4	6 ± 6
TB	Temple Buttle Formation (36.3°N, 112.1°W)	Dm-u	100	153 ± 2	−16	2	8° ± 1°N	−3 ± 3	−12 ± 4
MF	Martin Formation (34.8°N, 112.2°W)	Dm-u	100	158 ± 2	−13	2	6° ± 1°N	2 ± 3	−17 ± 4
CD	Columbus and Delaware limestone (40°N, 83°W)	Dm	100	164 ± 2	4	2	2° ± 1°S	−1 ± 3	−5 ± 4
CR	Columbus and Raisin R. dolomite (40°N, 83°W)	Dm	0	122 ± 9	−1*	12	0° ± 5°	−3 ± 6	36 ± 10
ON	Onondaga Limestone overprint (42.3°N, 74.3°W)	D1-m	100	169 ± 6	14	8	7° ± 5°S	4 ± 6	−3 ± 7
CA	Catskill Formation, flat-lying (42.0°N, 74.5°W)	Dm-u	100	172 ± 4	1	5	1° ± 2°S	−3 ± 4	−7 ± 5
CB	Catskill Formation, folded (39.8°N, 78.3°W)	Dm-u	8	164 ± 4	9	5	4° ± 2°S	0 ± 4	−1 ± 5

The age column gives the age of the rock, which is not necessarily that of the magnetization; *R* is the percentage of samples with reversed magnetization; alpha is the radius of the circle of confidence ($P = 0.05$); PLAT is the paleolatitude and its error. This error has been obtained by calculating first the error in *I* using the method of *Demarest* [1983], and then the limits to PLAT. Because *I* does not vary linearly with latitude, the mean is not centered within these limits, but the asymmetry is unimportant for the present task, and the error is, for brevity, quoted as half the difference between the limits. RPD and PR are the relative paleolatitudinal displacement and relative rotation calculated relative to the early Kiaman latitude and meridians. The early Kiaman paleopole is estimated as an average of entries 72 to 85 in the work of *Irving and Irving* [1982, Table 1] which is 44°N, 126°E alpha(95) = 4°. The paleopoles are derived from rock units that range in age from about 295 to 270 Ma (Late Westphalian to Sakmarian inclusive) with a mean age of about 280 Ma (Permo-Carboniferous boundary). The errors in RPD and RR are calculated from the errors in declination and paleolatitude and include errors in the grid as well as errors in the observations. The references are as follows: SN, herein; WV, *Murthy* [1983*a*]; DD, *Murthy* [1983*b*]; SH and SR, herein; DH, *Irving and Strong* [1984], SJ, *Scott* [1979]; TH, *Kent* [1982]; TB and MF, *Elston and Bressler* [1977]; CD and CR, *Martin* [1975]; CA, *Kent and Opdyke* [1978]; CB, *Van der Voo et al.* [1979].

boniferous paleolatitude of the Avalon Zone agrees with paleolatitude results from elsewhere in Acadia. Acadian results available from rocks in this age range are from the Upper Devonian Perry Formation of New Brunswick and the Belchertown Pluton of Massachusetts. *Black* [1964] studied volcanic units of the Perry Formation and obtained a mean direction of 184°, 34° alpha(95) = 15°. He used a "blanket" method of cleaning, a procedure which cannot now be regarded as satisfactory. *Robertson et al.* [1968] obtained a mean direction of 175°, 23° alpha(95) = 15° from red sediments of the Perry Formation. They used thermal demagnetization, recognized the complex nature of the magnetic memory, and stressed the uncertain and unsatisfactory nature of their result. These directions (irrespective of sign) are not grossly different from the *N* magnetization of the St. Lawrence Granite (Table 2). Although the inclinations are shallower, the maximum inclinations within statistical limits, are 49° and 38°, respectively. Knowing, as we now do [*Roy and Morris*, 1983; *Irving and Strong*, 1984; *Strong and Irving*, 1983], the widespread and pervasive nature of Kiaman remagnetization affecting a very wide variety of rock types from granites to red beds, it is probable that some overprinting remains unremoved in these rocks, thus lowering their inclination. Certainly, when due recognization is given to the uncertainties in this early work stated by its authors, there are no physical grounds for supposing the magnetization of the Perry Formation and the St. Lawrence Granite to be different, or for invoking any large-scale relative movement between them.

The Belchertown Pluton, dated at 380 ± 5 Ma, yielded a paleolatitude uncorrected for tilt of 7° ± 0°S [*Ashwal and Hargraves*, 1977] very different from our results but not significantly different from the Kiaman paleofield. The authors stress the uncertainty of possible tilt corrections. Clearly the meaning of this result is not clear, and no tectonic conclusions are drawn from it.

We now make comparisons with cratonic North America. Paleolatitudes obtained from Upper Devonian and lowest Carboniferous rocks of the craton range from 8°N to 7°S (Table 4, Figure 17), much lower than the 35°S observed from southeastern Newfoundland. At first sight this indicates a large (over 3000 km) post-Devonian displacement of southeastern Newfoundland from the south relative to cratonic North America. The sense of displacement is consistent with the Kent-Opdyke hypothesis, but the magnitude is about twice that invoked by them. But do the cratonic data truly record the Middle to Late Devonian cratonic field? In particular, are Kiaman overprints present? That they are may be suspected for two reasons. First, the near equatorial paleolatitudes observed in cratonic rocks are precisely those to be expected from Kiaman overprinting. Second, widespread Kiaman overprinting of the Carboniferous and Devonian record is known to have occurred in the Appalachians [*Kent*, 1982; *Irving and Strong*, 1984], it has been documented in some cratonic rocks [*Kent*, 1979], and it is especially well known in western Europe where it was first discovered [*Chamalaun and Creer*, 1964]. Indeed it is characteristic of the Paleozoic record [*Creer*, 1968].

The question can be partly answered by comparing the observations from Upper Devonian and Lower Carboniferous rocks with the early Kiaman paleofield. This is done by using RPD's and RR's (Table 4, Figure 17). These are the relative paleolatitudinal displacement (RPD) and relative rotation (RR) that the observing locality would have undergone if the magnetizations are indeed Kiaman. Insignificant (large) values indicate that the magnetization is (not) likely to have been much affected by Kiaman overprinting. For example, the

RPD for the N magnetization of the St. Lawrence Pluton exceed 30°, indicating that little or no Kiaman overprinting is present. This is to be expected because a Kiaman overprint (the H magnetization) has been separated experimentally (Table 4, entry SH) so the underlying N is likely to be "clean." In contrast, RPD's and RR's for the best studied cratonic unit (the Catskill Formation, CA, CB) are not significant, indicating that its magnetization could well be Kiaman.

RPD's for all the cratonic data are all insignificant, the observed inclinations in Upper Devonian and lowest Carboniferous rocks agreeing very well with the Kiaman paleofield. The rough concordance of directions in Upper Devonian rocks has often been noted in individual studies [e.g., *Kent and Opdyke*, 1978]. What Table 4 and Figure 17 show, for the first time, is that the effect is continent-wide. Five (TB, MF, CD, CR, CA) of the seven results yield significant RR's, but two of them (CD, CA) are marginal, only 1° outside error. The remaining three apparently are significant. The results TB and MF (Table 4) of *Elston and Bressler* [1977] both indicate clockwise rotation, and it is possible that the Colorado Plateau, where the localities are situated, has been rotated as a marginal consequence of the general dextral shear of the western cordilleran [*Beck*, 1976]. The third significant RR (MF) is based on data from only one site (10 specimens) and may be caused by inadequate sampling. The directions observed in Middle to Upper Devonian and earliest Carboniferous rocks are thus in general agreement with the Kiaman paleofield, and the inclinations (the most important tectonic parameter) are in perfect agreement. It is especially notable that those cratonic results that are based on the most detailed work (CA, CB) fit the Kiaman paleofield perfectly, both in declination and inclination.

Is there any internal evidence of Kiaman overprinting in the cratonic data itself? We suggest that there is. Infrequent normal magnetizations have been reported from folded parts of the Catskill Group (92% are reversed) and have low inclinations (−3° to −22°). These have been obtained by vector subtraction [*Van de Voo et al.*, 1979]. At temperatures of 650°–670°C, however, the inclinations in normally magnetized samples are higher, ranging from about −40 to −50° [*Van der Voo et al.*, 1979, Figure 3], and the directions show quasi-random variations, perhaps indicating an end-point, perhaps indicating that a magnetization of intermediate inclination is being isolated at these high temperatures. More detailed studies are urgently required to establish whether this is true or not, but on the basis of present evidence, the possibility exists that the true primary normal magnetization for the Catskill Group, fully freed of Kiaman overprinting, may have had an inclination (−40° to −50°) not dissimilar from that observed from southeastern Newfoundland.

We now consider polarities. The Kiaman paleofield is predominantly but not entirely reversed. Short intervals of normal polarity occur within it. *Irving and Pullaiah* [1976] calculate that during the Kiaman the paleofield was normal for only about 5% of the time. The record from Middle and Upper Devonian and Tournaisian rocks from cratonic North America corresponds closely. Of the eight units studied, two contain normal as well as reversed magnetizations and the others are predominantly reversed. In terms of specimens studied (total about 400) about 97% are reversed. Of course, it is possible that the Middle Devonian to Tournaisian paleofield was biased toward reversed polarity, like that of the Kiaman. This seems unlikely; for instance, it is not our experience in the St. Lawrence Granite which is entirely normal, and *Chamalaun and Creer* [1964] in their classical study of the Old Red Sandstone found only normal polarities amongst what they regarded as original magnetizations. Finally, we note that the normal and reversed magnetizations observed in the Ohio carbonates (entries CD and CR, Table 4) are not 180° apart. The cause of this is obscure, but it is possible that the normal and reversed magnetizations are of very different ages. It seems to us therefore, on the basis of present evidence, that the rare normal magnetizations observed in Middle and Upper Devonian rocks of the North American craton are suspect. Even if they are accurate records of the paleofield, they occur in a proportion which is consonant with their being Kiaman not Devonian.

Of the cratonic studies, two contain a fold test [*Graham*, 1949]. For the Onondaga limestone of New York the test is negative and it was magnetized after Alleghanian folding, which is late Carboniferous or Permian in age [*Kent*, 1979]. Since the magnetization is along the Kiaman field (Table 4) it is likely to be of this age, and no Devonian record is retained [*Kent*, 1979]. In contrast the folded Catskill Group of Pennsylvania, Maryland, and West Virginia the magnetization, predominantly or entirely, predates the Alleghanian folding [*Van der Voo et al.*, 1979]. The fact that the magnetization is prefolding has been used to argue that the magnetization is "not much later than Late Devonian and certainly pre-late Carboniferous" [*Van der Voo et al.*, 1979]. While this could be true, it is also possible that the magnetization was acquired during burial [*Hodych et al.*, 1984] or during the early stages of uplift following deep burial and immediately prior to folding. Both magnetization and folding could have occurred within the lengthy Kiaman interval. We suggest that the coincidence of the magnetization with the early Kiaman paleofield strongly supports this possibility (Figure 17).

Overprints observed from Newfoundland are now considered. Their paleolatitudes and rotations are shown in the inset to Figure 17 and listed in Table 4. All are in general agreement with the Kiaman paleofield. Overprints from the Deer Lake Group (DH) and from the Burin Peninsula (SR, SH) are consistent with a small anticlockwise rotation of Newfoundland, but the Terenceville overprint (TH, *Kent* [1982]) is slightly clockwise as is the southerly magnetization of Deadman's Bay dikes (DD, *Murthy* [1983*b*]). These are small divergences which could, in part, be experimental. Further study is required to determine their real cause.

Finally, we note that paleomagnetic results are available from Upper Devonian–Early Carboniferous plutons of the Piedmont of the southeastern United States [*Ellwood et al.*, 1980; *Ellwood and Abrams*, 1982; *Dooley*, 1983; *Barton and Brown*, 1983]. All have uncertainties concerning tilt corrections (see the exhaustive study of *Ellwood et al.* [1980]), and the Piedmont may or may not then have been attached to the North American craton [*Ellwood*, 1982]. Hence the results, although of much interest, are not germane to the Acadian problem, which is our present concern.

The observational picture therefore seems clear. The inclinations in cratonic Middle to Upper Devonian and lowest Carboniferous rocks are consistent with the early Kiaman paleofield and with Kiaman overprints observed from Newfoundland. Sometimes declinations show small statistically significant departures (whose possible meanings are discussed above), but data from the Catskill red beds and the Onondaga limestone, which are the best studied of the cratonic rocks, are totally consistent with the Kiaman paleofield. Hence there are no secure grounds on which to establish the Middle to Late

Devonian and earliest Carboniferous reference paleofield for cratonic North America. This picture is essentially identical to that described by *Roy and Morris* [1983] and *Irving and Strong* [1984] for the later Early Carboniferous (Viséan and Namurian). Observations from Upper Devonian and Lower Carboniferous rocks from Newfoundland and the Maritime Provinces, even though they have been freed of Kiaman overprinting, cannot therefore be used to determine whether or not these regions have moved relative to the craton. They can be so used once a reasonably secure cratonic reference paleofield becomes available. Consequently, our attempt to detect motions between the crator and the Avalon zone has not been successful, not necessarily because they have not occurred, but because the lack of accurate reference paleofields for cratonic North America does not allow the necessary comparisons to be made. Our new results, however, will provide the means of making such comparisons once reliable cratonic reference data for the Late Devonian and earliest Carboniferous become available.

It would seem therefore that the Kent-Opdyke hypothesis does not now rest very securely, because measurements across the Appalachian orogen itself in Newfoundland show that such motions did not occur in the Viséan or later [*Strong and Irving*, 1983; *Irving and Strong*, 1984], and because the Early Carboniferous [*Roy and Morris*, 1983; *Strong and Irving*, 1983; *Irving and Strong*, 1984; *Kent and Opydke*, 1984] and the Middle and Late Devonian (herein) record from the craton has probably been extensively overprinted about 50 Ma later in the Kiaman. This does not mean that motions of tens or even hundreds of kilometers, much less than those invoked by Kent and Opdyke, did not occur.

It is of interest to compare the history of development of ideas concerning possible Late Paleozoic displacements in the Appalachians with those of Mesozoic and Cenozoic displacements in the Western Cordillera of North America. In some early studies in the Cordillera, absence of motions was argued because widespread Cretaceous and Tertiary overprints were mistaken for original magnetizations (e.g., Guichon Batholith [*Symons*, 1971, 1983] or the Island Intrusions of Vancouver Island [*Symons*, 1970; *Yole and Irving*, 1980]. In the Appalachian discussion it is the cratonic data that now seem to have been commonly misinterpreted. Perhaps because of their apparently "unaltered" nature the seeming simplicity of their magnetizations and the variety of rock types studied, widespread overprinting of cratonic rocks has been considered unlikely, whereas in Acadia, partly because of the occurrence there of suitable igneous lithologies, and partly because studies there usually have been more detailed, the complex nature of magnetizations has been more generally recognized. The consequence has been that in Appalachian studies, ideas of displacement have been favored, and the alternative explanation, that the cratonic record is systematically younger than that in coeval rocks of the orogen, although considered, has not gained wide acceptance. The failure to recognize overprinting within the Cordillera lead, wrongly, to fixist interpretations. We suggest that the failure to recognize overprinting in the craton has lead, in Appalachian studies, to mobilist interpretations of the Late Paleozoic history which are no longer justified.

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